

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	45 m s ⁻¹
Total 1 mark			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses fact that at max speed driving force equals resistance	AO3.4	M1	$F = 30 \times 40$ $= 1200$
	States or uses $P = Fv$	AO1.2	B1	$P = (30 \times 40) \times 40$
	Obtains correct value for power	AO1.1b	A1	$= 48000 \text{ W}$
(b)	Uses resistance model in a three term equation of motion.	AO3.4	M1	$F - 30 \times 25 = 1200a$ $F = 1200a + 750$
	Obtains a correct equation of motion.	AO1.1b	A1	$48000 = 25(1200a + 750)$
	Solves 'their' equation of motion for a .	AO1.1a	M1	$a = \frac{1920 - 750}{1200}$
	Obtains correct acceleration. FT 'their' equation provided both M1 marks awarded	AO1.1b	A1F	$= 0.975 \text{ m s}^{-2}$ $= 0.98 \text{ m s}^{-2} \text{ to 2 sf}$
				Total 7 marks

Q3.

Using power = force $\times$ velocity

$$240\,000 = F \times 20$$

M1A1

$$F = 12\,000$$

A1

Accelerating force is $12\,000 - 5000$
 $= 7000\text{ N}$

B1

Using $F = ma$

$$22\,000a = 7000$$

M1

$$a = 0.318 \text{ or } \frac{7}{22} \text{ m s}^{-2}$$

A1

[6]

Q4.

- (a) using power = force $\times$ velocity
 power = $(25 \times 42) \times 42$

M1

$\therefore$ power is 44 100 watts

A1

2

- (b) when speed is 15 m s^{-1}

$$\frac{44100}{15}$$

 max force exerted is
 $= 2940\text{N}$

B1

resistance force is $25 \times 15 = 375\text{N}$
 accelerating force is $2940 - 375\text{N}$
 $= 2565$

M1

using $F = ma$

$$2565 = 1500a$$

m1

$$a = 1.71 \text{ m s}^{-2}$$

A1

4

[6]

Q5.

- (a) PE is $400 \times g \times 8 = 3200\text{ g}$ [or 31 360]

B1

1

(b) KE is $\frac{1}{2} \times 400 \times 2^2 = 800$

B1

1

(c) Work done per minute is 32 160 J

$$\text{Power} = 32\,160 \div 60$$

$$[(a) + (b)] \div 60$$

M1

$$= 536 \text{ W}$$

CAO Accept 537 from 31 400 in (a)

A1

2

[4]

Q6.

(a) $90 \text{ km h}^{-1} = 90 \times \frac{1000}{3600} \text{ ms}^{-1}$

$$= 25 \text{ m s}^{-1} \quad \text{AG}$$

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$$\text{B1: Must see } \frac{1000}{3600} \text{ or } \frac{1000}{60^2}.$$

B1

1

(b) Resistance is 5000 N

B1: Obtaining 5000.

B1

Using power = force $\times$ velocity

$$= 5000 \times 25$$

M1: Using $P = Fv$ with 25 and their F .

M1

$$= 125 \text{ kW}$$

A1: Correct final answer, must be in kW.

125W or 125 000 W B1M1

125 B1M1A1

A1

3

[4]

Q7.

- (a) At maximum speed,
tractive force = resistance force

M1

Using power = force $\times$ velocity:
 $800\,000 = F \times 40$

M1

$$F = 20\,000\text{ N}$$

A1

3

- (b) Using force $\times$ distance = work done = change in energy:

$$20\,000\text{ s} = \frac{1}{2} \times 60\,000 \times (40^2 - 36^2)$$

M1 $Fs = \text{change of KE}$

M1

A1 2 of 3 terms correct

A1

A1 all 3 terms correct

A1

Distance = 456 m

A1

4

[7]

Q8.

- (a) Power of engine is 8kW

$$\therefore \text{Force exerted by engine} = \frac{8000}{v}$$

M1 for Power = Fv

M1A1

Using $F = ma$:

m1

$$\frac{8000}{v} - kv^2 = 600 \frac{dv}{dt}$$

$$600 \frac{dv}{dt} - \frac{8000}{v} + kv^2 = 0$$

AG

A1

4

- (b) (i) When engine is turned off, power is zero:

$$-kv^2 = 600 \frac{dv}{dt}$$

AG

B1

1

(ii) $\int 600 \frac{dv}{v^2} = -\int k \, dt$

$$-\frac{600}{v} = -kt + c$$

Need '+ c'

M1

A1

When $t = 0$, $v = 20$:

$$\therefore c = -\frac{600}{20} = -30$$

A1

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$$\therefore \frac{600}{v} = kt + 30$$

When $v = 10$, $kt = 30$:

M1

$$\therefore t = \frac{30}{k}$$

$$-\frac{30}{k} \text{ SC3}$$

A1

5

[10]

Q9.

Max speed $\equiv$ zero acceleration used

Implied

M1

$$\frac{72000}{60}$$

M1

$$\frac{72000}{60} = k \times 60$$

$$k = 20$$

A1

[3]

Q10.

(a) Using Power = Force $\times$ Velocity:

M1

$$\text{Power} = 35 \times 50 \times 50$$

B1 for force 35 $\times$ 50

M1B1

$$= 87500 \text{ watts}$$

AG

A1

4

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(b) When speed is 30 m s⁻¹, resistance force is 35 $\times$ 30
= 1050 N

B1 for 35 $\times$ 30

B1

Force exerted by the engine is $\frac{87500}{30}$

M1

$$= 2916.7$$

Accept 2920, 2917 etc

A1

Using $F = ma$:

$$2916.7 - 1050 = 1500a$$

*At least 1 LHS term correct
(2 terms on LHS)*

M1

$$a = 1.24 \text{ m s}^{-2}$$

A1

5

[9]



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Examiner reports

Q1.

This question was very successfully answered, with 94% of students choosing the correct answer. No student chose 90 m s^{-1} , whilst the other two incorrect options were chosen equally often by the remaining students.

Q3.

Most students knew many of the techniques required to answer this problem. However many of them had not understood the theory behind these techniques. They tried to use specific formulae which could answer similar questions and frequently omitted minus signs. The safest approach was to find the driving force exerted by the engine and then to take away the resistance force, this gave the accelerating force on the train.

Q4.

In part (a), the majority of candidates showed that the power was $25 \times 42 \times 42$, giving the answer of 44 100 watts. Part (b) proved more demanding, with many failing to find the accelerating force to be 2940 (the maximum force exerted by the engine) – 375 (the resistance force).

Q5.

Most candidates completed parts (a) and (b) of this question correctly. In part (c), a number of candidates tried to use power = force $\times$ velocity and multiplied by 2, the speed of the water, instead of dividing by 60, the time in seconds.

Q6.

This question was usually answered well. In part (a), some just wrote down the result given, ignoring the requirement to 'show'. In part (b), a few left the power as 125 000 W instead of 125 kW.

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Q7.

Most candidates answered part (a) correctly. In part (b), a significant number did not read the question carefully to see that an *energy* method was required. Thus the *only* acceptable method was to use $Fs = \text{change in kinetic energy}$.

Q8.

The answer given in part (a) enabled virtually all candidates to create a solution, but a considerable minority of these were not convincing. It was necessary to include a

$$\frac{8000}{v} \text{ N}$$

statement that the force exerted by the engine was $\frac{8000}{v}$. Part (b)(i) was answered well, and in part (b)(ii), most candidates appreciated that the differential equation needed to be solved by separating the variables. Many candidates found the resulting integral

$$\int \frac{dv}{v^2}$$

challenging and the $+c$ term required was often omitted. For those who did use

limits, $\int_{10}^{20} \frac{dv}{v^2}$ was often found rather than $\int_{20}^{10} \frac{dv}{v^2}$.

Q9.

Most candidates completed part this question convincingly.

Q10.

Nearly all candidates quickly found the power to be 87 500 watts. In part (b), most found the resistance force to be 1050 N but only the better candidates realised that the force exerted by the engine had changed because the speed had changed.



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